IOTAFLOW
SYSTEMS PVT. LTD

Vortex Flowmeter Quick Guide

" If You Want To Manage It, Measure It "

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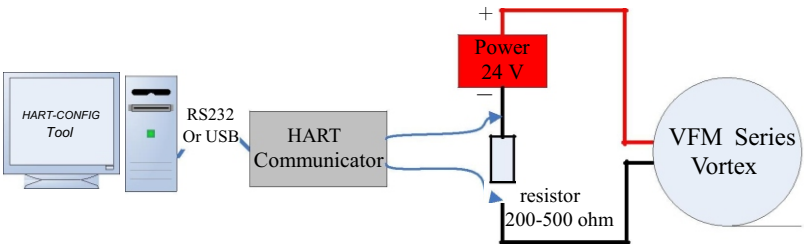
Step1: Prepare Configuration System



1. Connect the transmitter and power on.
2. Make sure the resistor is 200-500 ohm.
3. Run HART–CONFIG TOOL.exe software.
4. Set COM.
5. Use the left button of mouse to click the "Connect" -> "Search Address 0". All the configuration software user interface will be activated if communication is successful.

Step2: Set Vortex Parameters

1） Click on ‘Vortex Feature ’ -> ‘Vortex Parameters’ to enter the interface.
2） You can set meter size 【52】, fluid type【53】, low flow limit 【54】, high flow limit 【55】, maximum gain【56】, k-Factor 【57】, pulse factor unit 【58】and pulse factor【59】.
3） Click on ‘Write’ to save data.

Note：

In the HART-CONFIG Tool, select ‘Meter size and fluid type’ will automatically bring up the remaining parameters default values.
If the default value does not match the actual needs, each of which can be manually modified.

1. Channel select: Generally do not change!
2. Low flow limit: The unit is fixed at m^3/h. This value and the k-factor are used to determine the lower limit of the measured frequency.
3. High flow limit: The unit is fixed at m^3/h.
4. Maximum gain: Set maximum gain corresponding minimum frequency.
5. K-Factor: Set average calibration k-Factor.

Modify ‘H880_CH.INI’, you can change the default values for parameters. This file is in the installation directory.

Note: Low flow limit setting affects signal amplification. Please set low flow limit and high flow limit according to the actual flow limit!

Step2: Set Vortex Feature

1. Click on ‘Vortex Feature ’ -> ‘Vortex Feature’ to enter the interface.]
2. You can set flow mode, media status such as density, pressure, temperature, etc.
3. You can also reset the total flow.

1. Flow mode

Liquid Qv： Liquid volume
Liquid Qm： Liquid mass
Gas Qv： Gas volume：
Gas Qm： Gas mass：
Steam Qv： Steam volume
Steam(P/T)： Steam mass
Sat_Steam(T)： Saturated steam mas
temperature compensation
Sat_Steam(P)： Saturated steam mas
pressure compensation

Internal calculations with state transitions. When used in gas measurement, if you need to know the standard volumetric flow, the state must be set to:
Pressure: 0.0KPa
Temperature: 20℃

Special Note: The flowmeter has gas state transition function and steam density table. The standard volumetric flow and mass flow will changed if standard density, pressure or temperature changed.

2. Status

Gas: Set density, pressure, temperature. Liquid: Set density.

Step3: Configuration

1. Click on the ‘Transmitter Configuration’ -> ‘Range’ to enter into transmitter range configuration interface. Primary variable unit and upper range value can be selected or set.
2. Click on the ‘Transmitter Configuration’ -> ‘Output Setting’ to enter into the output parameters configuration interface. The damping value and display variables can be set.
3. Click on the ‘Transmitter Configuration’ -> ‘Fail Setting’ to enter into the alarm parameters configuration interface. The write protect setting , low alarm value and high alarm value can be set.

Step4: User Trim --- Five-point Linearity Correction

The actual k-Factor of vortex flowmeter is different in low flow rates and high flowrates. In order to improve the accuracy of vortex flowmeter, it provides 2 to 5 points k-Factor correction.

User trim steps：

1. Click on ‘User Trim’ -> “User Trim’ to enter into the interface.
2. Select ‘Calibration Points’.
3. Input ‘Frequency’ and ‘K Correction Factor’.
4. Click on the “Write Trim Data’ button to save the data to flowmeter.
5. Click on the ‘Cancel Trim Data’ will cancel the trim.

For example, for D = 80mm, measuring medium is liquid, the real k-Factor in different flowrates as follows:

<20 Hz	40	80	> 100
2200	2100	2100	2000

Then we can choose 4-points calibrated, set k-Factor 2100. Enter the calibration data as follows:

Frequency	k-Factor coefficient	*formula*
20	0.954545	2100/2200=0.954545
40	1	2100/2100=1
80	1	2100/2100=1
100	1.05	2100/2000=1.05

Step5: Additional Features

This operation is not required, should be determined according to the actual needs !

Low flow cutoff value：

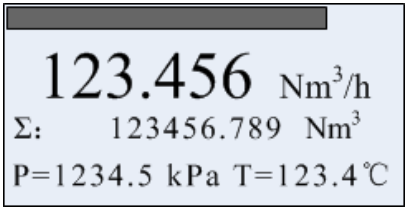
1. If the flow is smaller than the set percentage value, the flow will be set to 0.

Data Backup and Restore：

1. Enter ‘Advanced Features’ -> ‘Additional Features’, and click ‘Data Backup’ button to backup unit, range, damping, density, alarm setting, etc. (Only through HART-Config Tool to perform.)
2. Use HART handheld device like HART375 to set the damping to 5678, which will restore factory setting. (Note: Because 5678 is not a valid damping value, so returns' communication failure' when write damping is normal and does not affect data recovery.)
3. Refers local adjustment’s ‘Damping setting’, input ‘05678’ will restore factory setting. (Note: This operation does not affect the real damping value.)

LCD Display:

Three-Line Display Mode：

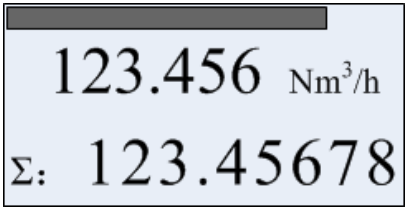


Displays the current percentage in progress bar.
First line display Instantaneous flow.
The second line display totalized flow.

The third line display frequency, pressure, temperature, density, current or percentage.

Short press M to set the third line shows the frequency, pressure, temperature, density, current, or percentages.

Two-Line Display Mode：



First line display Instantaneous flow.

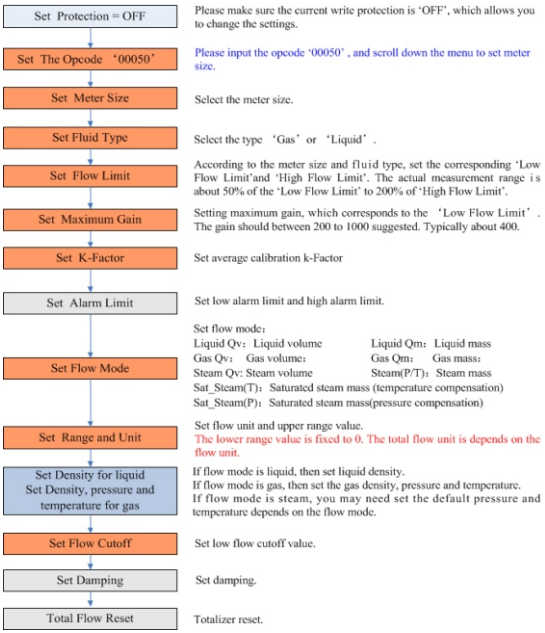
The second line display totalized flow.

Production Process via Local Adjustment

For a more detailed description of local adjustment operations, see User manual.

This color means that these items must be done.

This color means that these items must be done, and easily forgotten or incorrectly set.



Flow Mode Settings:

1. Gas Qv : Gas volume

A. Actual volume measurement, setting as follows:

Density: The density @ 20 ℃（Do not use.）
Pressure: 0.0KPa
Temperature: 20℃

B. Standard volumetric measurement, setting as follows:

Density: The density @ 20 ℃（Do not use.）
Pressure: Working pressure, gauge pressure KPa
Temperature: Working temperature, ℃

2. Gas Qm : Gas mass

A. Known current density, setting as follows: (The status conversion factor is 1)

Density: Working density (Current density)
Pressure: 0.0KPa
Temperature: 20℃

B. Known standard conditions density, setting as follows:

(According to the temperature and pressure for state conversion calculations)

Density: The density @ 20 ℃
Pressure: Working pressure, gauge pressure KPa
Temperature: Working temperature, ℃

Flow Mode Settings:

3. Liquid Qv : Liquid volume

A. Actual volume measurement, setting as follows:

Density: 1.0 or working density（Do not use.）

4. Liquid Qm : Liquid mass

A. Setting working density, setting as follows:

Density: Working density.

5. Steam Qv : Steam volume

Density: 1.0 or working density（Do not use.）
Pressure: 0.0KPa
Temperature: 20℃

6. Steam(P/T)、 Sat_Steam(T)、 Sat_Steam(P): Steam mass

A. Steam(P/T): (If external sensor fault, the input temperature or pressure is used for calculated density)

Pressure: Working pressure, gauge pressure KPa
Temperature: Working temperature, ℃

B. Sat_Steam(T): (If external temperature sensor fault, the input temperature is used for calculated density)

Temperature:Working temperature, ℃

C. Sat_Steam(P): (If external pressure sensor fault, the input pressure is used for calculated density)

Pressure: Working pressure, gauge pressure KPa

Recommended Settings Table Notes:

Note1: The local adjustment menu 53, 54, 55 and 56 is very important for vortex working well. Please carefully set according to the actual application.

Note2: H880 actual measuring range is from about 50% of the Low Flow Limit to 200% of the High Flow Limit setting. We recommended the Low Flow Limit and High Flow Limit setting range is not greater than 1:20 (1:10 is better). Maximum gain setting range: 40 to 1600.

Note3: H880 can work in 0.5 ~ 4000Hz input range. But the Low Flow Limit and High Flow Limit settings should be adjustment according the actual working conditions. In particular, it is used in the precession vortex, plug-in vortex, or steam measurements.

Note4: Set the value of 59 to 10 times of 57, making the output pulse frequency of 10 times the original pulse. It makes more easy calibration for large meter size.

Note5: For the use of the built-in vortex probe, if used for the liquid abd the lower limit can not be measured, choose 53 = GAS to amplify the signal. If the vortex sensor sensitivity is low, it can be such settings.

Assembly and calibration:

Note 1: The Vortex piezoelectric sensor line should be cut short when connection circuit board, and away from the power supply and the pulse output lines. The lines should not intertwined, otherwise will increases 50Hz interference.

Note 2: For H880TH or TBR, PT100 temperature sensor can be used instead of Pt1000 (line resistance effect becomes larger).

H880 Products model series:

H880WJ: 4~20mA with HART, Pulse output.

H880TH: 4~20mA with HART, Pulse output. Support external temperature and pressure sensor.

H880BR: Battery-powered, or 4-wire with RS485. Work time up to 3 years with a ER34615.

H880TBR: Battery-powered, or 4-wire with RS485. Support external temperature and pressure sensor. Work time up to 2 years with a ER34615.

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